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|-----------------------------------------------------------------------------------------------------------------------------|-----------------------|
|  <b>MOTION IMAGERY<br/>STANDARDS BOARD</b> | <b>MISB ST 1601.2</b> |
| <b>STANDARD</b>                                                                                                             |                       |
| <b>Geo-Registration Local Set</b>                                                                                           | <b>02 March 2023</b>  |

## 1 Scope

This standard defines metadata to support the identification of a geo-registration algorithm and various outputs from a geo-registration process. The intent is to use this standard in conjunction with the KLV construct Amend Local Set as defined in MISB ST 1607. Only include the Geo-Registration Local Set (LS) within an existing Local Set that contains items describing a sensor model.

## 2 References

- [1] MISB ST 0601.19 UAS Datalink Local Set, Mar 2023.
- [2] MISB ST 0902.8 Motion Imagery Sensor Minimum Metadata Set, Nov 2018.
- [3] MISB ST 1607.1 Constructs to Amend/Segment KLV Metadata, Mar 2023.
- [4] MISB MISP-2023.2: Motion Imagery Handbook, Mar 2023.
- [5] MISB ST 0107.5 KLV Metadata in Motion Imagery, Oct 2021.
- [6] SMPTE ST 336:2017 Data Encoding Protocol Using Key-Length-Value.
- [7] MISB ST 0807.26 MISB KLV Metadata Registry, Jun 2022.
- [8] MISB ST 1303.2 Multi-Dimensional Array Pack, Jun 2020.
- [9] ISO/IEC 9834-8:2014 Information technology - Procedures for the Operation of Object Identifier Registration Authorities - Part 8: Generation of Universally Unique Identifiers (UUIDs) and their use in Object Identifiers.
- [10] IETF RFC 4122 A Universally Unique IDentifier (UUID) URN Namespace, Jul 2005.
- [11] NGA.RP.0001\_1.0.0 NGA Recommended Practice for Universally Unique Identifiers, Jan 2013.

## 3 Revision History

| Revision  | Date       | Summary of Changes                                                                                                                                                                 |
|-----------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ST 1601.2 | 03/02/2023 | <ul style="list-style-type: none"> <li>Added Section 7 on using the Amend LS and the Metadata Substream Identifier (MSID)</li> <li>Revised Figure 3 to include the MSID</li> </ul> |

|  |  |                                                                                                                                                                             |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <ul style="list-style-type: none"> <li>Removed Universal Labels (Keys) for consistency with current MISB guidance</li> <li>Added acronyms and updated references</li> </ul> |
|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 4 Acronyms and Definitions

|                 |                                                                     |
|-----------------|---------------------------------------------------------------------|
| <b>KLV</b>      | Key Length Value                                                    |
| <b>LS</b>       | Local Set                                                           |
| <b>IETF</b>     | Internet Engineering Task Force                                     |
| <b>IEC</b>      | International Electrotechnical Commission                           |
| <b>ISO</b>      | International Organization for Standardization                      |
| <b>MDARRAY</b>  | Multi-Dimensional Array                                             |
| <b>MISB</b>     | Motion Imagery Standards Board                                      |
| <b>MISP</b>     | Motion Imagery Standards Profile                                    |
| <b>MSID</b>     | Metadata Substream Identifier                                       |
| <b>NGA</b>      | National Geospatial-Intelligence Agency                             |
| <b>RFC</b>      | Request for Comments                                                |
| <b>RP</b>       | Recommended Practice                                                |
| <b>SDCC-FLP</b> | Standard Deviation and Correlation Coefficient Floating Length Pack |
| <b>SMPTE</b>    | Society of Motion Picture & Television Engineers                    |
| <b>ST</b>       | Standard                                                            |

## 5 Introduction

Geo-registration, generically defined and as used in this document, is the process of revising sensor model metadata through a mathematical process. For example, a First Image may be geo-registered to a “known good” Second Image, and the mathematical process “transfers” the accuracy of the Second Image to the First Image by adjusting the sensor model metadata of the First Image. In another use case, typical in mosaic generation, two images “float” and adjustments to both sensor model’s metadata produce best alignment.

The basic metadata parameters needed for a sensor model are contained in an existing Local Set (e.g., MISB ST 0601 [1]) and profiled in MISB ST 0902 [2]. The KLV construct of the Amend Local Set, defined in MISB ST 1607 [3], enables revisions to the basic sensor model parameter values for subsequent geo-registration processes. The Motion Imagery Handbook [4] provides additional information on amending data of a parent metadata set.

The Geo-Registration Local Set provides the additional metadata necessary to identify geo-registration algorithms and additional geo-registration process output data not contained in the original LS.

## 6 Geo-Registration Local Set

The Geo-Registration Local Set is a KLV Local Set whose items provide information specific to a geo-registration algorithm, such as name and version, and additional geo-registration output, such as correspondence points. All KLV metadata needs to conform to MISB ST 0107 [5] and SMPTE 336 [6] encoding rules.

### 6.1 Conventions

| Requirement(s) |                                                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------------------------------|
| ST 1601-01     | All metadata shall be expressed in accordance with MISB ST 0107.                                                                 |
| ST 1601-02     | Formatting of Geo-Registration metadata shall be compliant with SMPTE ST 336 encoding rules for Universal Labels and Local Sets. |

### 6.2 Geo-Registration Local Set

The Geo-Registration Local Set 16-Byte Universal Label registered in MISB ST 0807 [7] is:

|                                                             |
|-------------------------------------------------------------|
| 06.0E.2B.34.02.0B.01.01.0E.01.03.03.01.00.00.00 (CRC 39238) |
|-------------------------------------------------------------|

Table 1 summarizes the Geo-Registration LS with columns defined as:

- **Tag ID** indicates the Tag to use when specifying the item in the Geo-Registration LS
- **Name** is the name of the item
- **Units** specifies the units of an item's value; a value of N/A means Not Applicable
- **Type** specifies the data type (binary format) when encoding the value into binary. The type uint is a generalization to allow the application to define the number of bytes needed for an item's value representation.
- **Rules** indicate if the item is Mandatory or Optional. The LS must include items listed as Mandatory, while those listed as Optional are at the implementer's discretion.

**Table 1: Geo-Registration Local Set Items**

| Tag ID | Name                                                                                                   | Units    | Type <sup>1</sup> | Rules     |
|--------|--------------------------------------------------------------------------------------------------------|----------|-------------------|-----------|
| 1      | Document Version                                                                                       | N/A      | uint              | Mandatory |
| 2      | Algorithm Name                                                                                         | N/A      | utf8              | Mandatory |
| 3      | Algorithm Version                                                                                      | N/A      | utf8              | Mandatory |
| 4      | Correspondence Points – Row / Column                                                                   | pixels   | MDARRAY           | Optional  |
| 5      | Correspondence Points – Latitude / Longitude                                                           | °        | MDARRAY           | Optional  |
| 6      | Second Image Name                                                                                      | N/A      | utf8              | Optional  |
| 7      | Algorithm Configuration Identifier                                                                     | N/A      | UUID              | Optional  |
| 8      | Correspondence Points – Elevation                                                                      | m        | MDARRAY           | Optional  |
| 9      | Correspondence Points – Row / Column Standard Deviation & Correlation Coefficients                     | variable | MDARRAY           | Optional  |
| 10     | Correspondence Points – Latitude / Longitude / Elevation Standard Deviation & Correlation Coefficients | variable | MDARRAY           | Optional  |

<sup>1</sup>Note on Lengths: Length computation occurs when the complete size of values is known at “run-time”. (E.g., The number of characters in the string determines the length of a utf8 string; likewise, the number of elements in the array determines the size of an MDARRAY. See the Motion Imagery Handbook for more information on data types and lengths.)

Items 4, 5, 8, 9 and 10 all contain information about a common list of tie points; therefore, the MDARRAYs for these items must have the same number of tie point elements.

| Requirement  |                                                                                                                                                                         |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ST 1601.1-03 | Where Geo-Registration Local Set Items 4, 5, 8, 9, or 10 are included in the Local Set, all the included items shall use the same number of tie points in their arrays. |

## 6.3 Geo-Registration Local Set Items

### 6.3.1 Item 1: Document Version

The Document Version item identifies the version of MISB ST 1601 used in the implementation.

| Requirement |                                                                                  |
|-------------|----------------------------------------------------------------------------------|
| ST 1601-04  | A Geo-Registration Local Set shall include the Document Version metadata Item 1. |

### 6.3.2 Item 2: Algorithm Name

The Algorithm Name item uniquely identifies the algorithm used to geo-register the imagery data to produce revised sensor model parameter values.

| Requirement |                                                                                |
|-------------|--------------------------------------------------------------------------------|
| ST 1601-05  | A Geo-Registration Local Set shall include the Algorithm Name metadata Item 2. |

### 6.3.3 Item 3: Algorithm Version

The Algorithm Version item is an alphanumeric that uniquely identifies the specific version of the Algorithm Name used to geo-register the imagery data.

| Requirement |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| ST 1601-06  | A Geo-Registration Local Set shall include the Algorithm Version metadata Item 3. |

### 6.3.4 Item 4: Correspondence Points – Row/Column

A geo-registration algorithm uses the Correspondence Points – Row/Column MDARRAY item to define an extensible list of tie points represented in pixel space for two Images, or a single Image and a set of geographic points defined in Item 5 (Latitude/Longitude) and Item 8 (Height above Elevation).

Figure 1 illustrates the first case of having correspondence or tie points between two Images: Image 1 and Image 2. The circles in Image 1 correspond to the triangles in Image 2. The MDARRAY stores each tie point's row and column.



**Figure 1: Illustration of Correspondence Points between two Images**

Table 2 shows the list of tie points for Figure 1, which becomes the MDARRAY.

**Table 2: Example Tie Points for Figure 1**

|         |        | Tie Points |     |     |     |
|---------|--------|------------|-----|-----|-----|
|         |        | 0          | 1   | 2   | 3   |
| Image 1 | Row    | 133        | 128 | 97  | 69  |
|         | Column | 31         | 91  | 122 | 129 |
| Image 2 | Row    | 89         | 82  | 52  | 27  |
|         | Column | 125        | 176 | 204 | 210 |

MISB ST 1303 [8] defines the MDARRAY KLV Pack construct.

The parameters for this Multi-Dimensional Array are: MDARRAY(Note<sub>A</sub>, 2, 4, Note<sub>B</sub>) where:

- Note<sub>A</sub>:

ST 1601 overrides the Data Type defined in MISB ST 0807 for the Row (CRC 44071) and Column (CRC 62839) items (shown below) from BER-OID to UINT to enable padding, as needed, to meet the MDARRAY requirement that all elements must be the same size.

Array(0, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.3F.01.00.00 (CRC 44071)<sup>a</sup>

Array(1, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.3F.02.00.00 (CRC 62839)<sup>b</sup>

Array(2, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.3F.01.00.00 (CRC 44071)<sup>c</sup>

Array(3, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.3F.02.00.00 (CRC 62839)<sup>d</sup>

tp = denotes a specific tie point

<sup>a</sup>This KLV key represents the row value of the tie point for the First Image.

<sup>b</sup>This KLV key represents the column value of the tie point for the First Image.

<sup>c</sup>This KLV key represents the row value of the tie point for the Second Image.

<sup>d</sup>This KLV key represents the column value of the tie point for the Second Image.

- Note<sub>B</sub>: This value is dependent upon the number of tie points produced between two Images.

This specifies a two-dimensional Array with dimensions of four (4) by that given by Note<sub>B</sub>.

When specifying tie points in pixel space to geographic points, use only one Image and specify the tie points in Item 5 for the latitude/longitude and Item 8 for the height above ellipsoid. Figure 2 illustrates a single Image with correspondences to geographic tie points.



| Geo Point | Latitude | Longitude | HAE  |
|-----------|----------|-----------|------|
| 0         | 32.98416 | 48.08388  | 1500 |
| 1         | 32.98417 | 48.08389  | 1501 |
| 2         | 32.98418 | 48.08390  | 1500 |
| 3         | 32.98419 | 48.08390  | 1499 |

**Figure 2: Illustration of Correspondence Points between an Image and geographic points**

Table 3 shows the list of tie points for Figure 2, which becomes three MDARRAYs. Item 4 specifies the Row/Column. Item 5 specifies the Latitude/Longitude. Item 8 specifies the Height Above Ellipsoid.

**Table 3: Example Tie Points for Figure 1**

|        |           | Tie Points |          |          |          |
|--------|-----------|------------|----------|----------|----------|
|        |           | 0          | 1        | 2        | 3        |
| Item 4 | Row       | 133        | 128      | 97       | 69       |
|        | Column    | 31         | 91       | 122      | 129      |
| Item 5 | Latitude  | 32.98416   | 32.98417 | 32.98418 | 32.98419 |
|        | Longitude | 48.08388   | 48.08389 | 48.08390 | 48.08391 |
| Item 8 | HAE       | 1500       | 1501     | 1500     | 1499     |

### 6.3.5 Item 5: Correspondence Points – Latitude / Longitude

A geo-registration algorithm uses the Correspondence Points – Latitude/Longitude MDARRAY item to define an extensible list of tie points represented in geographic space for two images. MISB ST 1303 [8] defines the MDARRAY KLV Pack construct.

The parameters for this Multi-Dimensional Array are: MDARRAY(Note<sub>A</sub>, 2, 2, Note<sub>B</sub>) where:

- Note<sub>A</sub>: Array(0, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.28.00.00.00 (CRC 53661)<sup>a</sup>  
 Array(1, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.29.00.00.00 (CRC 42793)<sup>b</sup>  
 tp = denotes a specific tie point

<sup>a</sup>This KLV key represents the latitude value of the tie point.

<sup>b</sup>This KLV key represents the longitude value of the tie point.

- Note<sub>B</sub>: This value is dependent upon the number of tie points produced between the two images.

This specifies a two-dimensional Array with dimensions of two (2) by that given by Note<sub>B</sub>.

### 6.3.6 Item 6: Second Image Name

The Second Image Name item uniquely identifies the second Image used in the geo-registration process.

### 6.3.7 Item 7: Algorithm Configuration Identifier

The Algorithm Configuration Identifier item is a generic identification system based on a Universal Unique Identifier (UUID) to differentiate configurations or set of parameters for a specific Algorithm Name and Algorithm Version. It is up to the algorithm vendor to identify the set of variable configuration parameters and values for each algorithm to form the basis of the UUID.

Multiple techniques to generate a UUID are available as discussed in ISO 9834 [9] and RFC 4122 [10]. NGA Recommended Practice [11] discusses allowed UUID versions.

This Standard recommends constructing UUIDs from unique configuration information through UUID Version 4 leveraging random numbers or UUID Version 5 leveraging Secure Hash

Algorithm (SHA) coding process. For example, combine minimization thresholds, minimum number of tie points, correlation settings, etc., and use either truly random numbers or the SHA hash function to generate the UUID.

### 6.3.8 Item 8: Correspondence Points – Elevation

A geo-registration algorithm uses the Correspondence Points – Elevation MDARRAY item to define an extensible list of elevation values for geographic space tie points for two Images. MISB ST 1303 defines the MDARRAY KLV Pack construct.

The parameters for this Multi-Dimensional Array are: MDARRAY(Note<sub>A</sub>, 1, Note<sub>B</sub>) where:

- Note<sub>A</sub>: Array(0, tp) = 06.0E.2B.34.01.01.01.01.0E.01.01.03.2A.00.00.00 (CRC 15605)<sup>a</sup>  
tp = denotes a specific tie point  
<sup>a</sup>This KLV key represents the elevation value of the tie point.
- Note<sub>B</sub>: This value is dependent upon the number of tie points produced between two Images.

This specifies a one-dimensional Array with a dimension of one (1) by that given by Note<sub>B</sub>.

### 6.3.9 Item 9: Correspondence Points – Row / Column Standard Deviation & Correlation Coefficients

A geo-registration algorithm uses the Correspondence Points – Row/Column Standard Deviation & Correlation Coefficients MDARRAY item to define an extensible list of standard deviations and correlations coefficients for tie points represented in pixel space for two Images. MISB ST 1303 defines the MDARRAY KLV Pack construct.

The value of Item 9 is a two-dimensional Array with dimensions of six (6) by the number of tie points (tp) defined in the value of Item 4, i.e., one sigma value for each value in Item 4's array and one rho value for each row/value pair.

The parameters for the Multi-Dimensional Array are: MDARRAY(Note<sub>A</sub>, 2, 6, Note<sub>B</sub>, E<sub>Bytes</sub> see Note<sub>C</sub>) where:

- Note<sub>A</sub>:  
 Array(0, tp) = Sigma value for Item 4's Array(0, tp), the First Image Row for tie point tp. Value is a floating point using IMAPB(0, 100, E<sub>Bytes</sub>)  
 Array(1, tp) = Sigma value for Item 4's Array(1, tp), the First Image Column for tie point tp. Value is a floating point using IMAPB(0, 100, E<sub>Bytes</sub>)  
 Array(2, tp) = Rho value for the cross correlation between the First Image row and column values for tie point tp. Value is a floating point using IMAPB(-1, 1, E<sub>Bytes</sub>)  
 Array(3, tp) = Sigma value for Item 4's Array(2, tp), the Second Image Row for tie point tp. Value is a floating point using IMAPB(0, 100, E<sub>Bytes</sub>)

Array(4, tp) = Sigma value for Item 4's Array(3, tp), the Second Image Column for tie point tp. Value is a floating point using IMAPB(0, 100, E<sub>Bytes</sub>)

Array(5, tp) = Rho value for the cross correlation between the Second Image row and column values for tie point tp. Value is a floating point using IMAPB(-1, 1, E<sub>Bytes</sub>)

tp = denotes a specific tie point

- Note<sub>B</sub>: This value is dependent upon the number of tie points produced between two Images.
- Note<sub>C</sub>: Compute the E<sub>Bytes</sub> value at runtime based on the given precision of the sigma and rho values. The E<sub>Bytes</sub> value is the maximum length across all IMAP'ed sigma and row values. For example, if a sigma values requires 3 bytes and the rho requires only 2 bytes then the rho values have their IMAP values computed with 3 bytes.

### 6.3.10 Item 10: Correspondence Points – Latitude / Longitude / Elevation Standard Deviation & Correlation Coefficients

A geo-registration algorithm uses the Correspondence Points – Latitude / Longitude / Elevation Standard Deviation & Correlation Coefficients MDARRAY item to define an extensible list of standard deviations & correlations coefficients for tie points represented in geographic space for two images. MISB ST 1303 defines the MDARRAY KLV Pack construct.

The Value of Item 10 is a two-dimensional Array with dimensions of up to six (6) by the number of tie points (tp) defined in Item 5's value, i.e., one sigma value for each value in Item 5's array and optionally Item 8's array, along with one rho value for each latitude/longitude pair in Item 5's array and rho values for the correlations between latitude/elevation and longitude/elevation.

The parameters for the Multi-Dimensional Array are: MDARRAY(Note<sub>A</sub>, 2, Note<sub>B</sub>, Note<sub>C</sub>, Note<sub>D</sub>) where:

- Note<sub>A</sub>:  
 Array(0, tp) = Sigma value in meters for Item 5's Array(0, tp), the latitude for tie point tp. Value is a floating point using IMAPB(0, 650, E<sub>Bytes</sub>). Note: Although the latitude is in degrees, in this standard measure the sigma value in meters.  
 Array(1, tp) = Sigma value in meters for Item 5's Array(1, tp), the longitude for tie point tp. Value is a floating point using IMAPB(0, 650, E<sub>Bytes</sub>). Note: Although the longitude is in degrees, in this standard measure the sigma value in meters.  
 Array(2, tp) = Rho value for the latitude and longitude cross correlation for tie point tp. Value is a floating point using IMAPB(-1, 1, E<sub>Bytes</sub>)  
 Array(3, tp) = Sigma value for Item 8's Array(0, tp), the elevation for tie point tp. Value is a floating point using IMAPB(0, 1000, E<sub>Bytes</sub>)  
 Array(4, tp) = Rho value for the latitude and elevation cross correlation for tie point tp. Value is a floating point using IMAPB(-1, 1, E<sub>Bytes</sub>)

Array(5, tp) = Rho value for the longitude and elevation cross correlation for tie point tp.  
Value is a floating point using IMAPB(-1, 1, E<sub>Bytes</sub>)

tp = denotes a specific tie point

- Note<sub>B</sub>: This value is three (3) if providing only latitudes / longitudes and is six (6) if providing latitudes / longitudes / elevations.
- Note<sub>C</sub>: This value is dependent upon the number of tie points produced between two Images.
- Note<sub>D</sub>: Compute the E<sub>Bytes</sub> value at runtime based on the given precision of the sigma and rho values. The E<sub>Bytes</sub> value is the maximum length across all IMAP'ed sigma and row values. For example, if a sigma values requires 3 bytes and the rho requires only 2 bytes then the rho values have their IMAP values computed with 3 bytes.

## 7 Usage of ST 1607

### 7.1 Metadata Substream Identifier (MSID)

A Metadata Substream Identifier (MSID) is a unique identifier within an Amend LS. This allows for multiple instances of Amend LSs within a metadata stream in support of sub-streams of sub-streams and Report-on-change. See the Motion Imagery Handbook for an in-depth discussion of sub-stream construction and the utility of the MSID. MISB ST 0601 defines requirements for the use of the MSID.

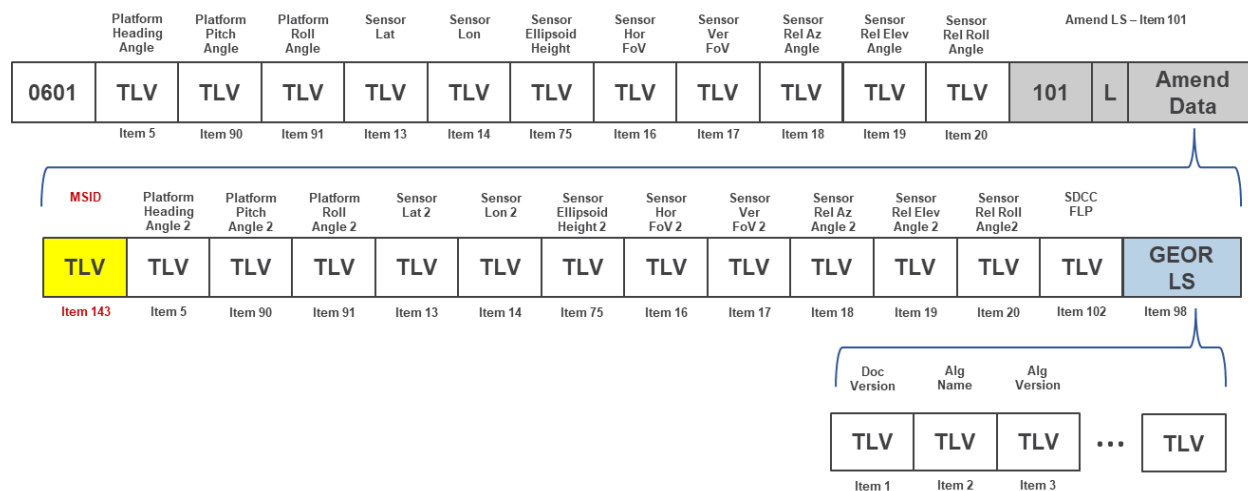
An instantiating metadata set defines a MSID metadata item and its requirements for its use. Every Amend LS implemented within that instantiating metadata set then includes that MSID as a metadata item.

## Appendix A – Example: Geo-Registration incorporated into MISB ST 0601 – Informative

In this example, a generic sensor model represented by 11 dynamic items from MISB ST 0601 define the imaging ray. Figure 3 illustrates the concept of how to leverage ST 0601 Item 101: Amend Local Set.

Item 101 reuses the ST 0601 Items 5, 90, 91, 13, 14, 75, 16, 17, 18, 19 & 20 with their values defined in the base ST 0601 LS. Item 101 also includes the MSID (Item 143) for unique identification of the Amend LS instance. The MSID together with the same ST 0601 11 items, with intended revised values, and with the addition of Item 102 (SDCC-FLP) reflect the results of a geo-registration process. Although the illustration only shows a sample of metadata items, the Amend Local Set Item 101 permits other ST 0601 items. A receiver is free to select items with their corresponding values from the base set or the “amended” substituted set.

In this illustration, Item 98 Geo-Registration LS (indicated as GEOR LS) is the final element in Amend LS Item 101, which includes specific attributes of the geo-registration process. Using multiple Item 101 Amend LSs within the ST 0601 LS enables the inclusion of multiple results from different geo-registration algorithms.



**Figure 3: Use of ST 0601 Item 101 - Amend LS embedding a Geo-Registration LS**